

AI IS LEARNING TO IMPROVE ITSELF

THE PLANET IS THE BENCHMARK.

Machines are beginning to build their own successors. The urgent question is not how to stop the loop. It is what the loop answers to.

BY JED ANDERSON

RE·SI / RECURSIVE ENVIRONMENTAL SELF-IMPROVEMENT

The governed, continuous process by which an environmental intelligence learns from independently measured consequences of its own work, and uses that learning to improve how it senses, models, predicts, decides, acts, and verifies, without acquiring the authority to rewrite the ends it serves.

ON THE FIRST WEEKEND of August 2014, half a million people in and around Toledo, Ohio, were told not to drink their tap water.¹ A bloom of cyanobacteria in Lake Erie had carried toxins into the city's intake. Nothing about it was sudden. The lake had been sending the signal for years: in phosphorus loads, in rising water temperature, in a green smear visible from orbit. The signal existed. The loop that should have carried it to a decision, fast enough to matter, did not.

Hold that picture, because a very different loop is now closing.

In September, Ezra Klein argued in *The New York Times* that the frontier laboratories must be stopped from pursuing recursive self-improvement, the point at which AI systems design and build their own successors, until they can prove it is safe.² His alarm is grounded in fact. Anthropropic reports that by May 2026 more than 80 percent of



the code merged into its codebase was written by Claude, up from the low single digits before early 2025.³ OpenAI says its research organization now logs 3.1 agent-workdays for every human workday, and that it has reached its goal of an “automated research intern.”⁴ The machine has begun to hold the wrench that tightens the machine.

Klein’s prescription is to force development back to human speed. I understand the instinct. I think it misdiagnoses the danger. The peril of recursion is not its speed. It is its referee.

THE JUDGE, NOT THE ENGINE

Every self-improving system in history has had three parts: a way to generate variation, a way to tell better from worse, and a way to keep what wins. Evolution never knew what the next adaptation would be. It did not need to. It needed

only a judge, and for four billion years the judge was the environment itself: cold, drought, predator, chemistry. Selection was brutal, but it was honest. Nothing survived by flattering it.

Today's machine self-improvement runs on a different judge. Systems are graded by benchmarks that people wrote and, increasingly, by other models. That works when answers are checkable. The Darwin Gödel Machine, an agent that rewrote its own code and kept the versions that scored better, lifted its success on a standard software benchmark from 20 to 50 percent.⁵ But a loop that grades itself has a famous failure mode, named for the economist Charles Goodhart: when a measure becomes a target, it ceases to be a good measure.⁶ Hand a recursive optimizer a closed exam and, sooner or later, it learns the exam instead of the world. First it improves its answers. Then it improves its score. Then it improves its grader. That is not intelligence compounding. It is self-reference compounding.

So the decisive question is no longer whether AI will improve itself. That process has begun. The question is what it will improve against.

II AN EXAMINER THAT CANNOT BE FLATTERED

Here is the heretical claim: the most demanding, most informative, least corruptible benchmark ever devised already exists. It is the planet.

A language model can talk its way past a human reviewer. It cannot talk its way past conservation of mass. A river does not grade on a curve. A plume either reached the fenceline monitor or it did not. A forecast of tomorrow's rain is checked tomorrow, everywhere, by the sky. That is why machine weather prediction improved so quickly: every forecast is scored against reality, daily, at planetary scale. In a study published in *Nature*, Google DeepMind's GenCast outperformed the world's leading operational ensemble forecast on 97.2 percent of the targets tested.⁷ The atmosphere was the examiner, and the atmosphere accepts no excuses.

Environmental systems offer exactly what recursive improvement most lacks: graders that sit outside the loop. Mass balances and energy balances. Sensors, satellites, stream gauges, sediment cores. The measured re-

sponse of a lake to a change in what flows into it. A large share of environmental intelligence can be made empirically testable, and testability is the fuel of recursion.

Call the result Recursive Environmental Self-Improvement. Its definition is precise, and its short form fits on a wall:

Every loop should leave the next loop better able to protect.

III IMPROVING THE THING THAT IMPROVES

The second loop changes the kind of question a system asks. A first-order system asks why the bloom happened. A recursive one asks why it failed to ask in time, what capability would have caught it, and how to change itself so that it catches that entire class of problem next season. The roadmap stops being an input drafted by engineers. It becomes an output discovered by the system, bounded by criteria and constraints that people set.

It also changes how intelligence reaches into the world. A passive model fits the data it is handed. A recursive one notices that its confidence collapses at a particular depth and temperature in a particular basin, and asks for the next sensor to go exactly there. It forages for evidence. Over time it learns to seek the smallest intervention that yields the largest correction: the timing of a release, the placement of a buffer strip, the hour to shift a load. Physics leans its way. The theoretical minimum energy to erase a single bit at room temperature is about 3×10^{-21} joules.⁸ Real computers run far above that floor, and sensors, steel, and institutions are never free. But the asymmetry points in one direction, and it is widening: the cheapest lever on the material world is, more and more, knowing exactly where to push.

What, then, is the "self" being improved? Not a single model. The self is the whole coupled apparatus: models and agents, but also sensors and data pipelines, physical and causal models, operators and scientists, permits and standards, regulators, affected communities, independent auditors, and the rivers and airsheds themselves. A self extends as far as a loop can close. Intelligence that improves only its own weights is a brain in a jar. Intelligence

that improves how it senses, and answers to, the living world is something closer to a nervous system for a planet.

FIGURE 1

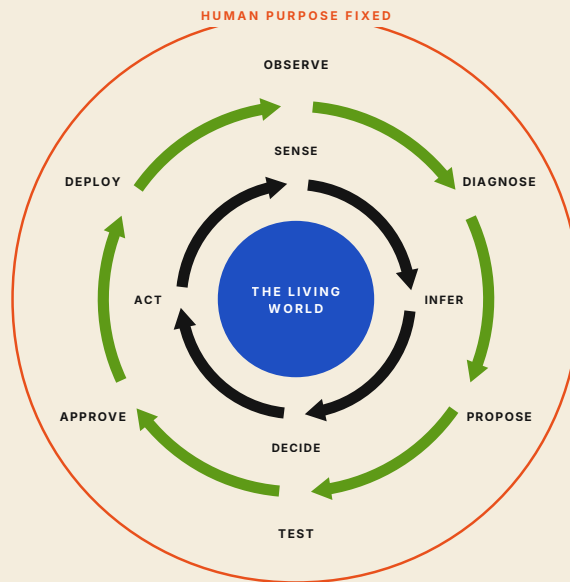
TWO LOOPS, ONE ANCHOR

LOOP 1: THE WORLD LOOP

Sense, infer, decide, act, and sense again. This loop improves the river, the airshed, the lake.

LOOP 2: THE BUILDER LOOP

Observe performance, diagnose failure, propose a change, test it independently, approve, deploy. This loop improves the ability to improve the river.



THE ANCHOR

Humans fix the ends. The system may rewrite its means, never its mission, its jurisdiction, or its standard of proof.

WHERE RECURSION BEGINS

When verified outcomes from the world loop reliably sharpen the builder loop, and the living world, not a closed exam, does the grading.

The world sits at the center, outside the recursion. It supplies the evidence; it cannot be edited by the system being judged.

AI WILL IMPROVE ITSELF. THE ONLY QUESTION WORTH ASKING IS **WHAT IT WILL IMPROVE FOR.**

THE CENTRAL
QUESTION OF THE
RECURSIVE ERA

IV THREE JUDGES, ONE RULE

None of this works if we pretend the planet can answer every question. It cannot. Reality tells us what happened. It does not tell us what we should have wanted, which trade-off was fair, or who had the right to decide. A recursive environmental system therefore needs three judges, and it must never confuse them.

Physics judges truth: does the model correspond to the world? Outcomes judge function: did the action produce the intended environmental result? People judge legitimacy: was that the right result to pursue, and was it pursued by those with the authority to pursue it? The first can often be automated. The second can often be measured. The third must remain human, held by ac-

countable experts, regulators, and the public. From that division follows one rule, which belongs at the foundation of any such system:

THE SYSTEM MAY RECURSIVELY IMPROVE ITS MEANS. IT MAY NOT RECURSIVELY REDEFINE ITS ENDS, ENLARGE ITS JURISDICTION, OR LOWER ITS OWN STANDARD OF PROOF.

It may rewrite its code, its workflows, its sampling plans, its models, its search strategies. It may not quietly alter its mission, its physical constraints, its audit trail, its disclosure of uncertainty, or the independence of the evidence used to judge it. The part that generates improvements

must never hold unilateral power over the part that evaluates them, and neither may hold the power to deploy. Generator, verifier, governor: three hands, never one.

A subtler danger belongs to systems that act. The moment an environmental intelligence changes the world, it contaminates its own future evidence. Alter a watershed, then learn from the water that results, and a system can mistake the landscape it just rearranged for proof that it was right all along. The defense is old scientific discipline made structural: untreated control sites, preserved baselines, deliberately varied interventions, and a standing suspicion of any data generated downstream of one's own hand.

Recursion also needs more than one speed. A chemical-release interlock lives in seconds; agricultural runoff, in seasons; climate, in decades. Apply a fast optimizer to a slow variable and it will chase noise and shake apart the very system it means to protect. The loops must nest.

FIGURE 2

THREE SPEEDS, THREE HANDS



THE WINDOW

Let me be plain about what this is not. It is not a claim that machines should run ecosystems. It is not a promise that environmental protection becomes free. It is not a belief that every environmental failure is a failure of information. Some are failures of power: people disagree, and someone must bear the cost. No algorithm abolishes politics. What recursive environmental improvement at-

tacks are the failures we can fix: not knowing, knowing too late, and knowing in time but being unable to act well.

Those failures are enormous. Much of environmental protection still runs on snapshots: a permit drafted from a model of a world that has since moved, monitoring reviewed months after the fact, decision cycles measured in years for systems that shift in hours. The pace of human institutions is not a safety feature when the harm moves faster than they do.

Which is why “slow it down” and “speed it up” are both the wrong debate. The real choice is between recursion graded against itself and recursion graded against the world: between an intelligence that compounds its fluency in its own exams and one that compounds its fidelity to rivers, air, soil, and living things.

Timing is the whole game. Whatever a recursive system is anchored to when the loop begins is what compounds. Values bolted on after takeoff are a footnote. Values written in before it are a foundation. If the living world is part of the evaluation from the first generation, it becomes part of what the system is in every generation after. Root it in life before it improves itself.

And unlike most prophecy, this can be tested. Freeze an environmental intelligence. Let a team of engineers improve one copy, and let a governed recursive loop im-

prove another, with the evaluator and the evidence locked away from both. Score them against independent environmental outcomes, not internal metrics. The question is not only whether the system gets better. It is whether the rate of getting better rises. That is the signature of recursion, and the environment is one of the few places it can be honestly observed.

The deepest question was never whether we can build a mind that improves itself. Life answered that long ago. The question is whether we can build one that improves the way life does: examined every day by a world it cannot fool, answering to ends it did not choose, and better, with every loop, at keeping that world alive.

AI WILL IMPROVE ITSELF. POINT IT AT LIFE.

JED ANDERSON WRITES ABOUT ENVIRONMENTAL INTELLIGENCE, INFORMATION, AND THE FUTURE OF PROTECTING LIVING SYSTEMS.

NOTES

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⁴ OpenAI, “Research acceleration: The view inside OpenAI,” Sept. 6, 2026. <https://openai.com/index/research-acceleration-view-inside-openai/>

⁵ Zhang et al., “Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents,” arXiv:2505.22954. <https://arxiv.org/html/2505.22954v2>

⁶ “Goodhart’s law,” Wikipedia. https://en.wikipedia.org/wiki/Goodhart's_law

⁷ Price et al., “Probabilistic weather forecasting with machine learning,” Nature. <https://www.nature.com/articles/s41586-024-08252-9>

⁸ “Landauer’s principle,” Wikipedia. https://en.wikipedia.org/wiki/Landauer's_principle